QR CODE **WATCH THE VIDEO BEFORE YOU SOLVE** YouTube • Lesson 1

RULE · Multi-step equations (ko'p qadamli tenglama)

Order: clear fractions → expand → collect the variable → combine like terms.

$$\frac{2}{3}(x+3) = x - 1 \qquad \leftarrow \text{clear the fraction } (\times 3)$$

$$\downarrow$$

$$2(x+3) = 3(x-1) \qquad \leftarrow \text{expand parentheses}$$

$$\downarrow$$

$$2x + 6 = 3x - 3 \qquad \leftarrow \text{collect the variable}$$

$$\downarrow$$

$$6 + 3 = 3x - 2x \qquad \leftarrow \text{combine like terms}$$

$$\downarrow$$

$$x = 9$$

WARNING · Check what's actually asked

Most missed SAT point: solving for x correctly, then answering x when a different expression is asked.

$3x + 5 = 20$. Find $6x + 10$.

Wrong: $x = 5$, answer "5".

$x = 5 \rightarrow 6(5) + 10 = 40$.

Right: 40, **not** x.

RULE · Sign-flip rule (ishora almashish qoidasi)

Operation	Sign flips?
Add or subtract	No
Multiply / divide by a positive	No
Multiply / divide by a negative	**Yes**

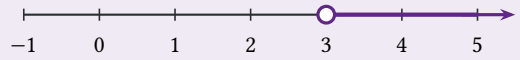
$-3x + 5 \leq 20 \;\Rightarrow\; -3x \leq 15 \;\Rightarrow\; x \geq -5$ $(\div(-3)$, sign flips)

RULES SHEET – 2/2

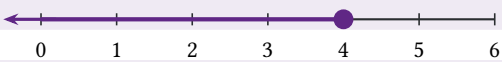
RULE · Graphing the solution

Sign	Circle	Meaning
$<$ or $>$	Open circle	Boundary excluded
$\leq$ or $\geq$	Closed circle	Boundary included

$x > 3$ **(open)**



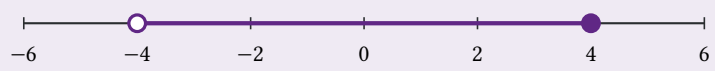
$x \leq 4$ **(closed)**



RULE · Compound inequalities (qo'shma tengsizlik)

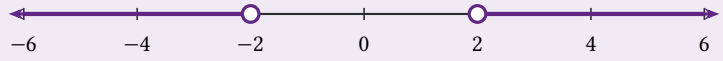
"**And**" – same operation on all three parts: $-3 < x + 1 \leq 5 \Rightarrow -4 < x \leq 4$

$-4 < x \leq 4$



"**Or**" – solve each part separately: $2x + 1 < -3$ or $2x + 1 > 7 \Rightarrow x < -2$ or $x > 3$

$x < -2$ **or** $x > 3$



WARNING: the operation must apply to **all three parts** – not just one.

RULE · Interval notation (oraliq belgilash)

Traditional textbooks use a semicolon; SAT/English sources use a comma – same meaning:

Traditional: $(2;\ 5]$ SAT: $(2,\ 5]$